Hooke sliding to rolling

LAMMPS dem05 linear_history | Tangential friction and spin | 28 October 2026

PASS	4 of 4 runs pass both official 1% endpoint velocity checks.

Official endpoint check at 0.75 s

Run	dt (us)	vx (m/s)	Error (%)	omega_y (rad/s)	Error (%)	1% checks
CPU dt	10	0.714236806	0.006847	285.761175	0.016411	PASS
GPU dt	10	0.714236805	0.006847	285.761175	0.016411	PASS
CPU dt/2	5	0.714237445	0.006758	285.760535	0.016187	PASS
GPU dt/2	5	0.714237445	0.006758	285.760535	0.016187	PASS

Targets: vx = 5/7 = 0.714285714286 m/s; omega_y = vx/R = 285.714285714 rad/s. The official slip_cessation checker requires each relative error to be at most 1%, at analytic_segment = 2 (step 75000).

Unchanged official inputs and mapping

Quantity	Official input / FunDEM configuration
Sphere / initial state	R = 0.0025 m; density 2700 kg/m^3; center z = 0.002499982 m; vx = 1 m/s; zero spin
Wall / gravity	Floor z = 0; upper plane z = 1 m, inactive; gravity = 1 m/s^2 downward
Contact law	Hooke kn = 10000 N/m; linear-history kt = 2800 N/m; sliding friction = 0.4
Damping / materials	Restitution 1; manual damping override off; automatic damping = 0. Explicit linear stiffness makes E/nu placeholders inactive.
Other terms	Rolling/torsional friction and global damping = 0; no fitted coefficients or solver changes
Time and output	0.75 s; dt = 10 / 5 us; 75,000 / 150,000 steps; 751 native frames per run at 1 ms; CSV every step

Ideal Coulomb solution and contact convention

I = (2/5) m R^2; vx = 1 - 0.4 t; omega_y = 400 t.
s_R = vx - R omega_y = 1 - 1.4 t; t* = 2/(7 mu g) = 0.714285714286 s.
After ideal slip cessation, vx = 5/7 and omega_y = (5/7)/R. The official check tests endpoint velocities, not t*.

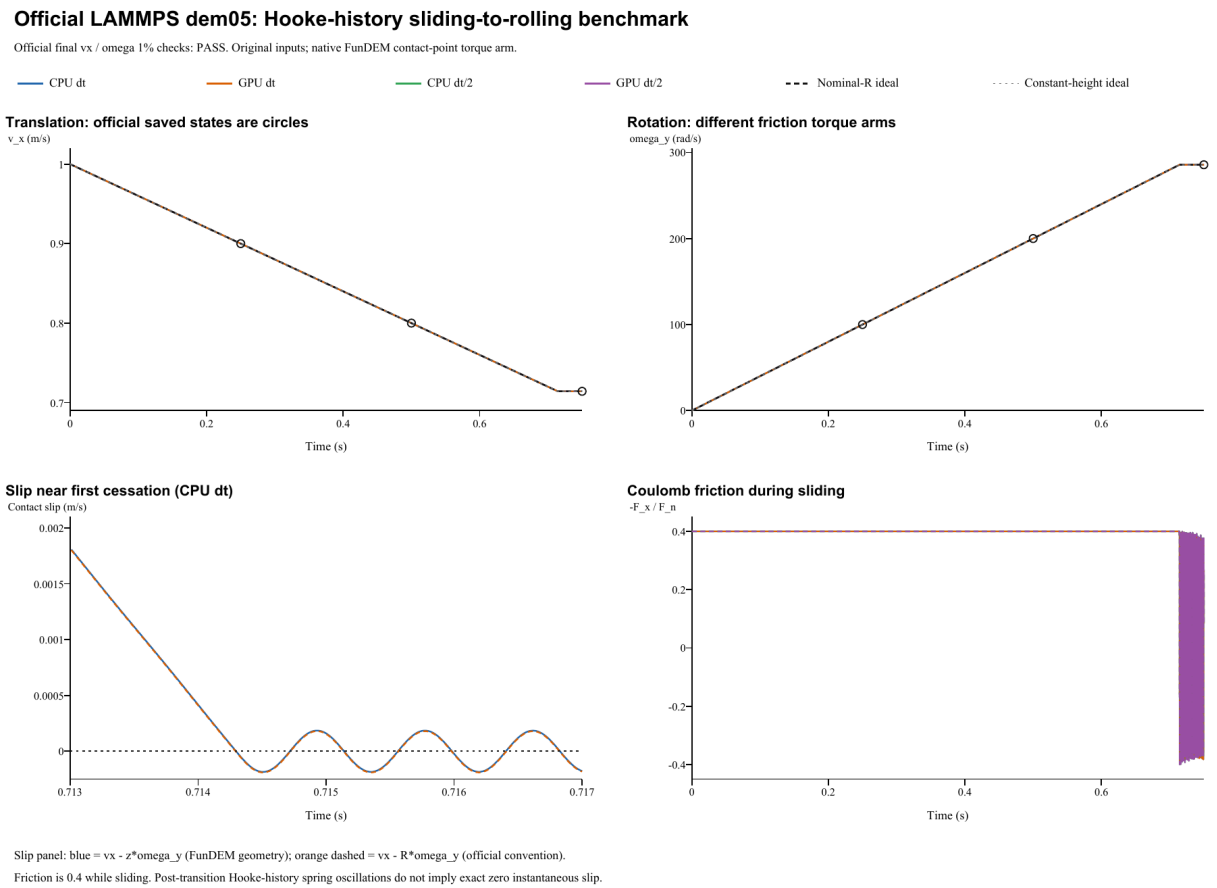
FunDEM uses a = R - delta = z for slip and friction torque; LAMMPS uses nominal R. Initial overlap is 18 nm (0.000720% of R). This difference is retained without radius fitting. Tangential damping is zero: an active viscous-force limiter is not tested.

Reference provenance and boundary mapping

Official YAML and saved states | Analytic checker. Commit f9720ce2d176d615495d1d6c31349b1f71be1c73; YAML blob 9d4f2bbc7cf3aa5c10728a2cb2e90cbce024fa91. LAMMPS was not rerun locally; YAML epsilon 1e-10 is not a cross-code tolerance. Unwrapped motion on the uniform plane is equivalent here to the isolated periodic sphere; golden positions are compared after x wrapping into [-0.05, 0.5] m.

Curves and retained evidence

Real native trajectories, ideal targets and pinned official saved states



Numerical and contact checks

Maximum CPU/GPU history differences: 1.96e-11 m/s in velocity and 1.96e-08 rad/s in spin. Halving dt changes the CPU endpoint vx by 6.4e-07 m/s and omega_y by 0.00064 rad/s.

Mean sliding abs(Ft)/Fn ranges from 0.40000000 to 0.40000000. Maximum torque-derived arm error = 4.34e-19 m. A first slip zero can precede tangential spring oscillations; it is not proof of sustained exact rolling.

Complete configured native results are retained

CPU dt: run.json | GPU dt: run.json | CPU dt/2: run.json | GPU dt/2: run.json

751 native frames per run, 3004 in total, including run.json, project.json, configured frames/contacts, loadable projects, audits and backend logs. Native output is every 1 ms (100 base steps / 200 half steps). Every integration step is separately recorded in CSV; native output was not every step.

Run manifest | Parameter mapping | Analysis JSON | Checkpoint CSV | Contact audit | Complete result ZIP

Execution and validation scope

The helper advances real FunDEM SimulationSession with linked 0.4.3 CPU/CUDA native libraries. No solver source or official physical parameter was changed. GUI executable byte identity was not established.

This Hooke variant has its own endpoint outcome and does not replace the earlier dem05 Mindlin failure. It exercises sliding friction, spin-up and approach to rolling. Zero tangential damping, rolling resistance and torsion leave their nonzero behavior untested; many-particle flow requires other cases.